Loka Kalyan Balla

Software Developer

Location: CA | **Phone:** (408) 381-7848 | balla.lk027@gmail.com | LinkedIn

Summary

- Developed scalable, responsive web applications using Django, Flask, FastAPI, Tornado, React.js, Node.js, Angular, and Bootstrap, leveraging 4+ years of experience in full-stack development.
- Possesses a diverse skill set spanning Python, Java, JavaScript, C++, React, TypeScript, and various front-end and back-end technologies.
- Designed and implemented RESTful APIs, GraphQL, and gRPC services, utilizing JSON and XML for efficient data exchange in microservice environments.
- Automated processes and created scripts with Python and Bash, enhancing efficiency through web scraping with BeautifulSoup and Scrapy.
- Architected microservices, monolithic applications, SOA, and distributed systems, applying SOLID principles, OOP, MVC, and Design Patterns to optimize performance.
- Deployed cloud infrastructures on AWS (EC2, S3, Lambda, RDS), Azure, and GCP, utilizing Docker, Kubernetes, Terraform, and CloudFormation for containerization and infrastructure automation.
- Led continuous integration and deployment pipelines with Jenkins, CircleCI, and GitLab CI, enforcing GitFlow strategies to enhance code quality and delivery.
- Managed relational (PostgreSQL, MySQL, SQLite) and NoSQL databases (MongoDB, Redis), optimizing queries and database performance through indexing, normalization, and stored procedures.
- Developed data-driven solutions using Pandas, NumPy, Matplotlib, and Scikit-learn, training machine learning models to solve complex analytical problems.
- Tested and debugged software using Unit Testing, TDD, BDD, and integration tools like pytest and PDB, ensuring code reliability and maintainability.
- Led Agile teams in Scrum and Kanban environments, facilitating sprint planning and cross-functional collaboration to deliver high-quality solutions.
- Mentored teams and designed scalable software architectures, enhancing system performance and reliability.

Technical Skills

Languages:	Python, JavaScript, C, C++, PHP, TypeScript, SQL.
Front End Technologies:	HTML, CSS, Bootstrap, REST APIs, MapReduce, Apache, Front-End, Design Patterns, Application Development.
Backend Technology:	Django, Flask, ReactJS, Micro Services, NumPy, Pandas, Fast API.
Methodologies:	Agile, SDLC & Waterfall.
Version Control Tools:	Git, GitHub.
Database:	Oracle, PostgreSQL, SQL Server, MySQL, MongoDB.
Cloud Technologies:	MS Azure, AWS, Code Pipeline, Code Build, Batch, EC2, ECS, SES, Lambda, DynamoDB, VPC, SQS, SNS, CI/CD.
Software Tools and OS:	Postman, RabbitMQ, Swagger, MacOS, Windows, Linux, Unity, Docker, VS Code, Airflow React Native, Deep, Learning, Kubernetes, Kafka, HBase, Spark, Unix, Yarn, Pytest and Unit test.

Professional Experience

W4M.ai.

Full Stack Developer **Jul 2024 - Present**

- Designed and developed a scalable full-stack web application with Python, FastAPI in the backend and React, TypeScript in the frontend, serving over 2,000 users, enhancing performance and user experience through efficient, responsive interactions.
- Restructured the codebase to a service-oriented architecture (SOA) using FastAPI, SQLAlchemy, and PostgreSQL, increasing modularity and scalability by 40%.
- Managed AWS infrastructure by configuring EC2, S3, and Lambda, and built CI/CD pipelines for automation, streamlining deployment processes and reducing manual intervention.
- Automated job description creation by integrating OpenAI's LLM API and Meta-Llama model, reducing manual effort by 80% and increasing speed by 50%, while ensuring reliability with a retry mechanism for generating accurate job titles and descriptions.

Novo Nordisk, US

Software Development Engineer **Sep 2023 - May 2024**

- Architected and built a collaborative simulation platform for the R&D team, enabling data scientists to test models efficiently, fostering collaboration and accelerating research and development cycles.
- Implemented over 90 automated unit tests using Pytest to streamline the testing process, significantly improving code coverage and ensuring better test reliability, contributing to more efficient bug detection and resolution.
- Refactored project modules utilizing pylint and flake8, improving code quality and maintainability for Python-based projects using NumPy, Pandas, and Matplotlib, which reduced technical debt and enforced consistent coding standards.
- Optimized and maintained CI/CD pipelines using Jenkins, GitLab, and Python-based scripting, adding regression tests to ensure stability in production environments and reducing deployment-related issues.
- Collaborated with cross-functional teams through Agile methodologies to gather requirements, design solutions, and deliver high-quality software, utilizing tools like Jira and Confluence while implementing Python REST APIs to meet business objectives and user needs.
- Enhanced user experience and interactivity in web applications by developing dynamic web pages using Python Flask for backend services and integrating AJAX for responsive front-end interactions.
- Integrated external APIs using Python Requests and Flask to fetch and display live data, enriching the application’s functionality and keeping the platform up-to-date with real-time information.
- Leveraged multi-threading and asynchronous programming in Python (via asyncio and threading) to improve performance in handling large data sets, enabling more efficient data processing and real-time updates.
- Implemented robust data validation and security practices in Python, including input sanitization and error handling, to ensure compliance with industry standards and safeguard user data.

Arizona State University, Tempe, AZ
GSA **Aug 2022 - May 2024**

- Engineered data pipelines with Pandas, SQL reducing processing time by 10 hours for datasets over 1 million records.
- Managed real-time processing of student actions, such as assignment submissions and quiz completions, using AWS Lambda for event-driven functions and DynamoDB for scalable, low-latency data storage, ensuring a seamless and responsive experience for over 100 students.
- Led the migration of legacy applications to a cloud-native architecture using AWS Lambda, improving system scalability, reducing operational overhead, and enhancing application performance.
- Refactored and optimized a legacy Python codebase, implementing design patterns and best practices to enhance system performance and reduce technical debt.
- Implemented comprehensive error handling and logging mechanisms using Python logging libraries, which minimized system downtime and facilitated faster issue resolution through improved traceability.
- Applied key Java concepts such as Collections, exception handling, and Java 8 features (lambda expressions, stream API, functional interfaces) to meet project requirements efficiently.

Societe Generale Global Solution Center, Bangalore, India
Software Engineer **Oct 2021 - Jul 2022**

- Designed a PaaS product for 500+ users with asynchronous RESTful APIs, reducing response time by 20%.
- Enhanced unit test coverage by taking ownership of code refactoring and maintenance, resulting in a more reliable and efficient codebase.
- Designed and implemented a microservices architecture utilizing Docker and Kubernetes, streamlining deployment processes and increasing scalability.
- Engineered real-time financial data streaming with Kafka and integrated Spark for high-performance analytics, significantly reducing data processing delays and optimizing throughput.
- Deployed a centralized logging system on AWS for microservices, improving the debugging process by reducing troubleshooting time and enhancing system reliability.
- Led cross-functional collaboration in Agile environments (Scrum, Kanban) to ensure timely project delivery, fostering a culture of continuous improvement and teamwork.

Trigent Software, Hyderabad, India
Software Developer **Oct 2020 - Sep 2021**

- Designed and developed an end-to-end automated invoice processing system, leveraging Python, Django, and OCR technology, resulting in enhanced data extraction and improved accuracy in financial workflows.
- Integrated the AIPS with existing ERP systems using RESTful APIs, facilitating seamless data transfer and reducing manual reconciliation errors across departments.
- Architected a scalable solution utilizing AWS EC2, S3, and Lambda functions, improving system performance and ensuring scalability to handle high-volume invoices.
- Implemented machine learning models using Scikit-learn to detect invoice anomalies, improving error detection and reducing manual reviews.
- Collaborated with cross-functional teams including finance and IT departments, demonstrating strong communication and teamwork skills to align project goals and system functionality with business requirements.
- Developed a robust invoice validation system using SQL and PostgreSQL, automating data checks and reducing inconsistencies in the validation process.
- Utilized Agile methodologies, participating in Scrum meetings and sprint planning, ensuring timely project delivery while improving team collaboration and project visibility.
- Optimized OCR algorithms using Tesseract and OpenCV, increasing document processing speed and enhancing data extraction accuracy for unstructured invoice data.
- Performed unit testing using PyTest and debugging tools, ensuring system reliability and reducing downtime by identifying and resolving critical bugs.
- Documented key processes and system architectures, supporting the finance team with detailed technical documentation, improving knowledge transfer and onboarding efficiency.

Education

Arizona State University, Tempe, AZ
Masters in Software Engineering **May 2024**

B.M.S Institute of Technology and Management, India
Bachelors in Computer Science and Engineering **Aug 2021**

Projects

Comparative Study & Analysis of Eye gaze estimation models, Capstone Project (ASU)

- Conducted a comparative study and analysis of four eye gaze tracking estimation models (iTracker, FAZE, GAZEL, ODABE), evaluating their suitability based on application requirements and exploring opportunities for performance enhancement.

Human Resource Management System | Tech stack: Java, MySQL

- The System encompasses Employee attendance and overtime monitoring, allowances, deductions, generating payslip.
- Utilized Role-Based Access Control (RBAC) mechanisms to enforce granular permissions enhancing UX within MVC architecture

Semantic Enrichment of Museum Tours with Linked Data | Tech-stack: SPARQL, React, Apache Fuseki and Python, Flask for backend

- Spearheaded the development of the Smart Museum Tour, to deliver a more enriching and seamless exploration of museum displays. Provided detailed information on artists, artworks, and associated areas of interest, enhancing the overall museum visitor experience. Tech-stack: SPARQL, React, Apache Fuseki and Python, Flask for backend